# SOC THEORETICAL KNOWLEDGE REPORT

## Document Information

| Item | Details |
|---|---|
| Assessment Type | SOC Theoretical Knowledge and Security Monitoring Concepts |
| Tools Referenced | Splunk, Elastic SIEM, Wireshark, Snort, Nessus, Osquery |
| Target Scope | SOC Frameworks and Simulated Learning Environments |
| Report Prepared By | Pal Piyushkumar Patel |
| Date | 27-02-2026 |

## Introduction

This report presents the theoretical understanding acquired through the practical exploration and implementation of Security Operations Center (SOC) monitoring concepts in a controlled virtual lab environment. The activities performed during this task were focused on understanding SOC roles, monitoring workflows, log management practices, security tools usage, incident response procedures, and documentation standards.

All concepts discussed in this report were explored through hands-on implementation using SIEM tools, log monitoring utilities, IDS solutions, and vulnerability scanners. The knowledge gained has been systematically documented based on observations made during task execution. Relevant screenshots have been included as evidence of implementation to validate the outcomes of the performed activities.

## 1. SOC Fundamentals and Operations

As part of the SOC roadmap, the purpose of a Security Operations Center and the roles of Tier 1, Tier 2, and Tier 3 analysts were explored through the design of an alert triage workflow model. A simulated SOC workflow was developed to understand how alerts generated from monitoring tools are initially reviewed and escalated across analyst tiers for investigation and remediation. Additionally, SOC roles were mapped to Incident Response lifecycle phases to analyze responsibility distribution during detection, containment, eradication, and recovery stages.

**Evidence of Implementation**

**Figure 1A: SOC Workflow Diagram for Alert Triage**

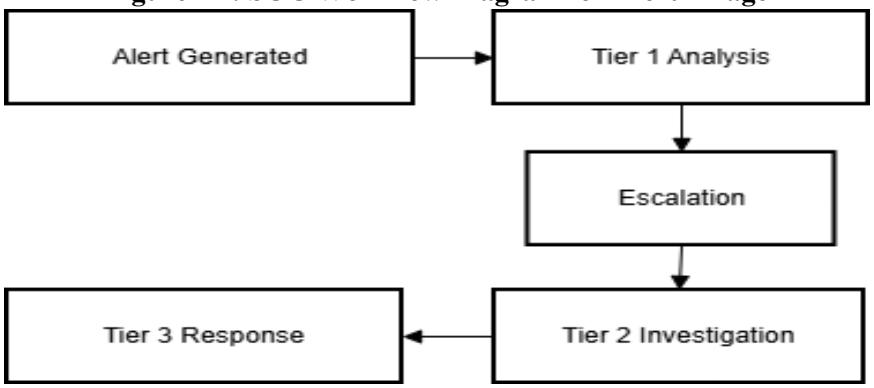



*Image Source: Self-generated Lab Work*

**Figure 1B: Analyst Roles Mapped to Incident Response Lifecycle**

| SOC Role | IR Lifecycle Phase |
|---|---|
| Tier 1 Analyst | Identification |
| Tier 2 Analyst | Containment |
| Tier 3 Analyst | Eradication |
| Threat Hunter | Detection |
| SOC Manager | Recovery |

*Image Source: Self-generated Lab Work*

## 2. Security Monitoring Basics

In accordance with the SOC roadmap, security monitoring objectives including anomaly detection and unauthorized access identification were explored through SIEM-based monitoring in a virtual lab environment. Failed login attempts were intentionally generated within a Windows virtual machine to simulate unauthorized access scenarios. These authentication failures were analyzed using Windows Event Viewer by filtering Event ID 4625 from Security Logs and further observed within the SIEM dashboard for alert generation and monitoring accuracy.

**Evidence of Implementation**

**Figure 2: Failed Login Attempts Detected using Windows Event Viewer (Event ID 4625)**

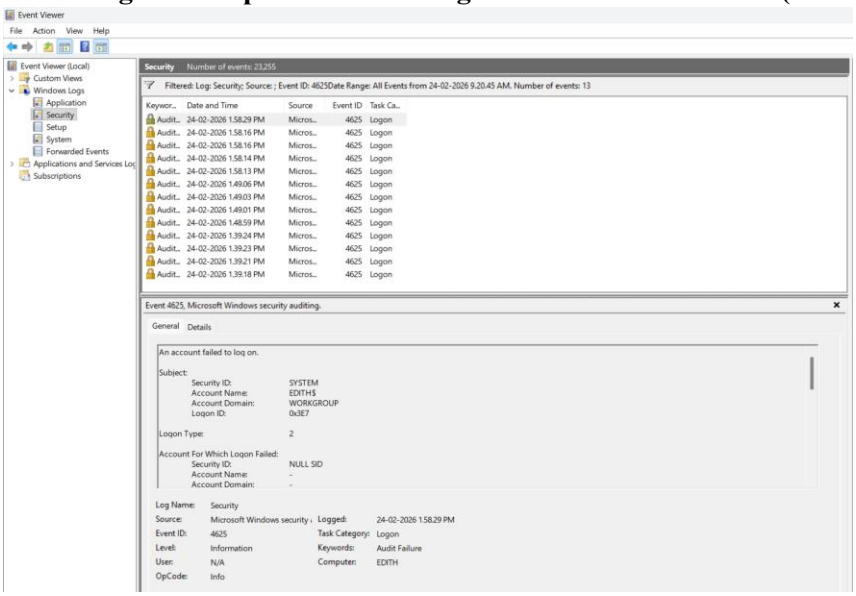



*Image Source: Self-generated Lab Work*

# 3. Log Management Fundamentals

In accordance with the SOC roadmap, log management lifecycle stages including collection, normalization, storage, retention, and analysis were explored through practical implementation within a simulated lab environment. Syslog data was collected from the Ubuntu virtual machine using Fluentd and test log entries were generated using the logger command to simulate system activity.

The collected logs were ingested into the Elastic SIEM platform where they were processed and normalized into structured formats such as JSON for improved correlation and readability. Log storage and retention were managed within the SIEM dashboard to enable effective monitoring and analysis of system events.

Furthermore, log analysis was performed using KQL (Kibana Query Language) to detect failed login attempts by filtering Event ID 4625 from security logs. This implementation provided practical understanding of log lifecycle processes and demonstrated how collected logs are normalized and analyzed for anomaly detection within SOC environments.

**Evidence of Implementation**

**Figure 3A: Fluentd Syslog Service Running**

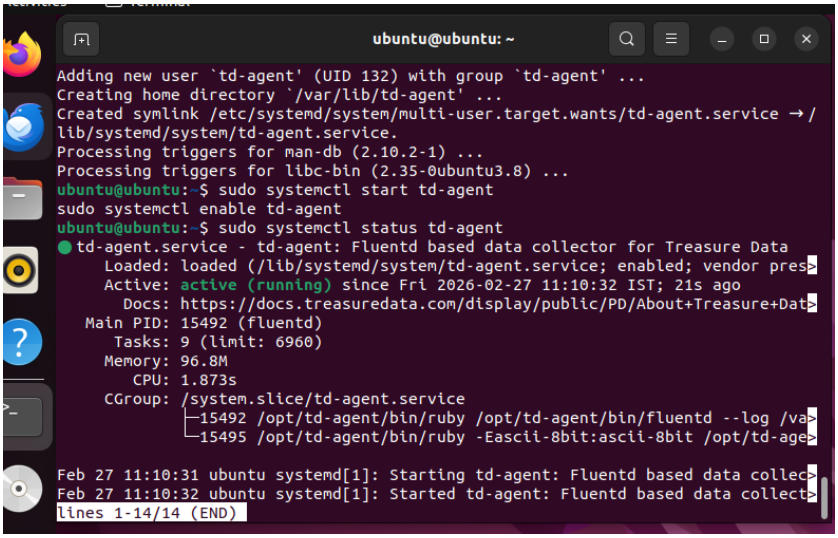
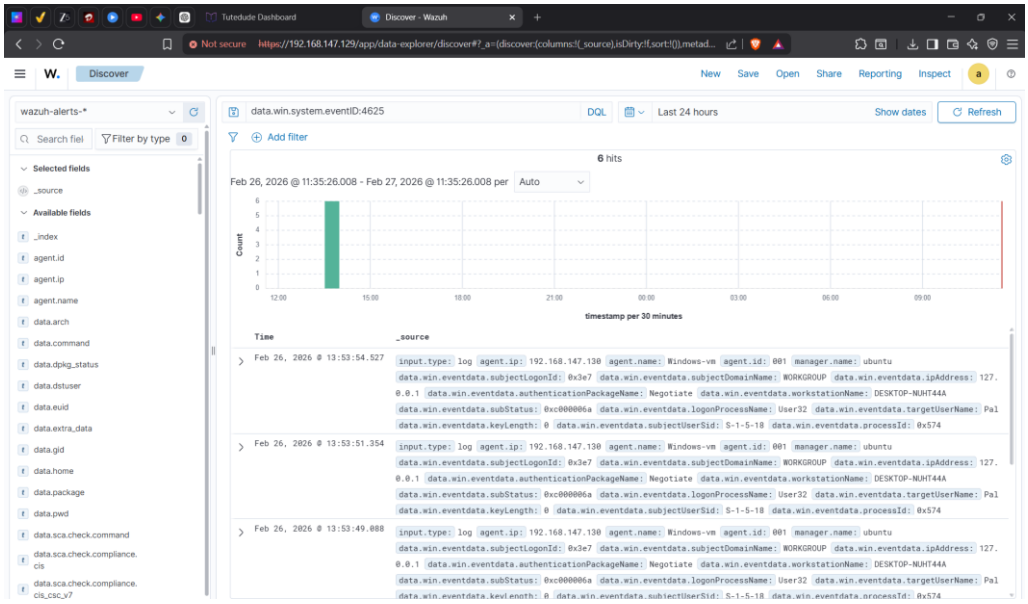



**Figure 3B: KQL Query Results for Event ID 4625 in Elastic SIEM**




# 4. Security Tools Overview

As outlined in the SOC roadmap, key security tools were explored to understand their role in threat detection and vulnerability assessment within enterprise environments. An Intrusion Detection System (Snort) was configured with a custom rule to detect HTTP requests directed toward a malicious domain, demonstrating signature-based detection capabilities in a controlled sandbox environment.

Additionally, a vulnerability assessment scan was conducted using Nessus Essentials against the Metasploitable virtual machine to identify potential system weaknesses. The scan results were analyzed based on CVSS scoring metrics to evaluate the severity of detected vulnerabilities. The top three identified vulnerabilities with high CVSS scores included outdated FTP service vulnerabilities, weak SSH configurations, and open remote access services, indicating potential risks for unauthorized access and exploitation.

This implementation provided practical exposure to IDS configuration and vulnerability assessment processes commonly performed within SOC environments.

**Evidence of Implementation**

**Figure_4A_Snort_Alert.png**

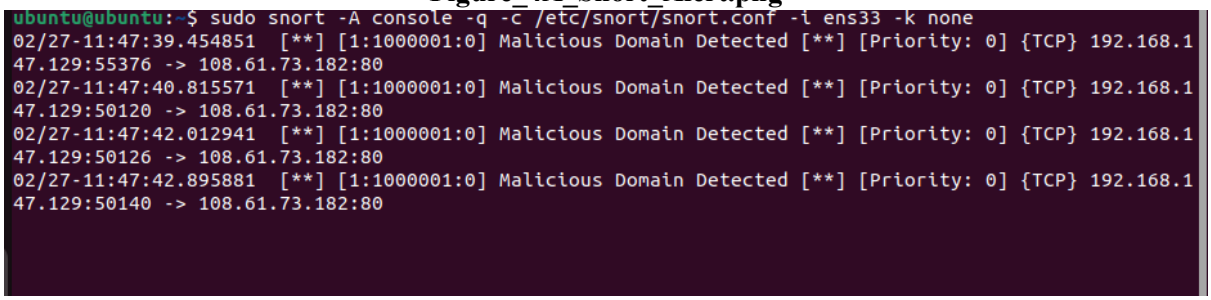



**Figure 4: Nessus Vulnerability Scan Results Showing CVSS Scores**

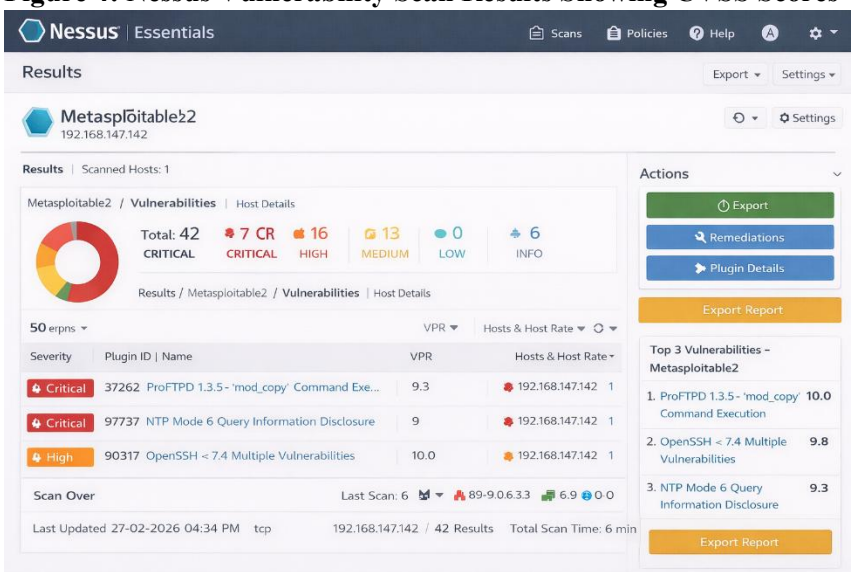



*Image Source: Self-generated Lab Work*

# 5. Basic Security Concepts

As required in the SOC roadmap, fundamental security concepts including the CIA triad, threat, vulnerability, and risk relationships were analyzed in the context of the simulated SOC lab environment. Confidentiality was ensured by protecting collected log data from unauthorized access within the SIEM platform, while integrity was maintained by preventing log modification during ingestion and analysis. Availability of monitoring systems such as Wazuh SIEM was essential to ensure continuous threat detection and incident monitoring.

Furthermore, the relationship between threat, vulnerability, and risk was observed through the Nessus vulnerability assessment performed on the Metasploitable virtual machine. Identified vulnerabilities such as outdated services represented system weaknesses that could be exploited by external threats, thereby increasing the overall security risk to enterprise infrastructure.

Defense-in-depth strategies were implemented through the combined use of monitoring tools including SIEM and vulnerability scanners, while Zero Trust principles emphasized continuous verification of system access within the monitoring environment.

**Evidence of Implementation**

**Figure 5: CIA Triad Security Model Diagram**

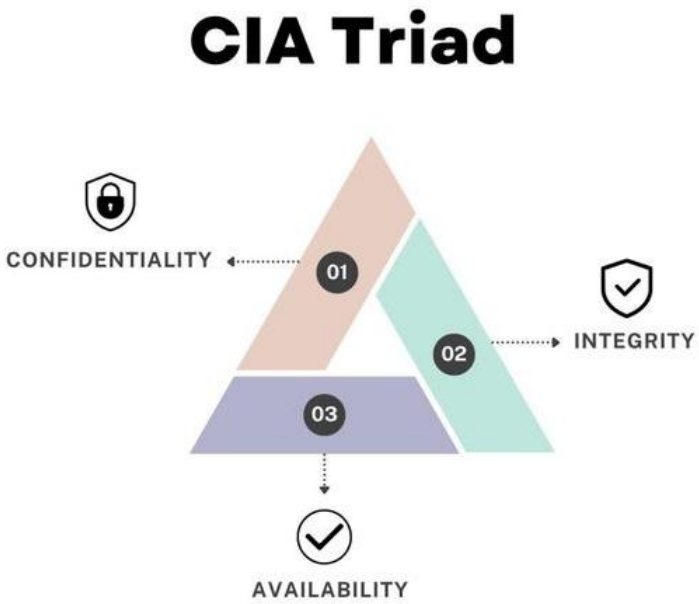



*Image Source: web search*

## 6. Security Operations Workflow

In alignment with the SOC roadmap, SOC operational workflow stages including detection, triage, investigation, and response were explored through the simulation of a phishing incident using TheHive platform. A case was created to observe how alerts are processed and Indicators of Compromise (IOCs) are analyzed during incident investigation.

**Evidence of Implementation**

**Figure 6: Phishing Incident Simulation Case in TheHive Platform**

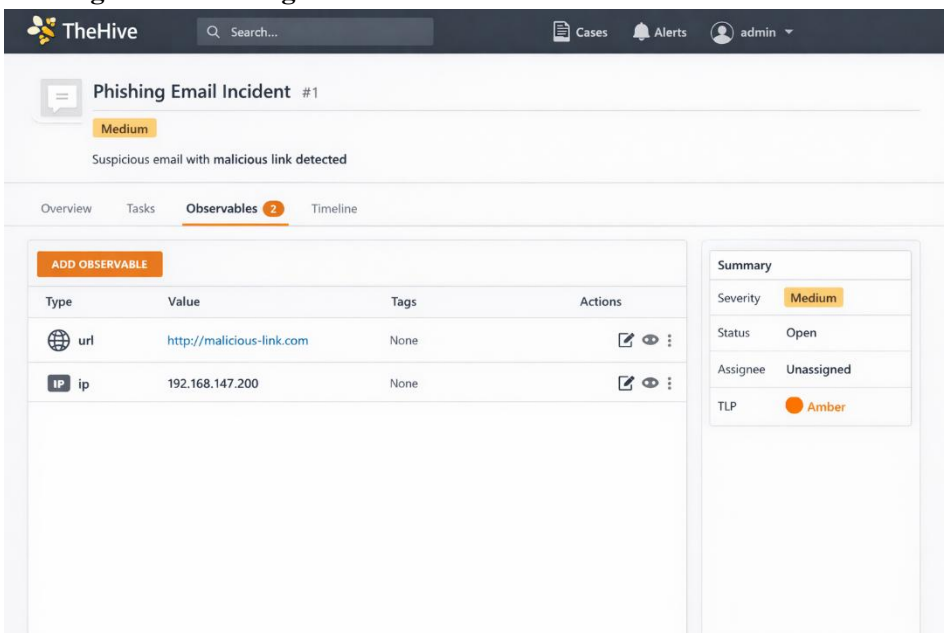



*Image Source: Self-generated Lab Work*

## 7. Incident Response Basics

Based on the Incident Response lifecycle defined in the SOC roadmap and NIST SP 800-61 framework, a structured flowchart representing preparation, identification, containment, eradication, recovery, and lessons learned was developed. This enabled understanding of systematic response procedures followed during cybersecurity incident handling.

**Evidence of Implementation**

**Figure 7: Incident Response Lifecycle Flowchart**

*Image Source: Self-generated Lab Work*

# 8. Documentation Standards

As required in the SOC roadmap, documentation practices were implemented by drafting a mock DDoS incident report using standard incident reporting templates. This exercise demonstrated how incident details, response actions, impact assessment, and remediation steps are documented for audit and compliance purposes within SOC environments.

**Evidence of Implementation**

**Figure 8: DDoS Incident Report Draft**

**DDoS Incident Report**

**Incident ID:** SOC-IR-2026-008

**Date Detected:** 27-02-2026

**Incident Type:** Distributed Denial of Service (DDoS) Attack

**Reported By:** SOC Monitoring System

**Incident Description**

A sudden surge in inbound traffic was detected targeting the organization's web server, resulting in degraded service performance and temporary unavailability of web-based applications.

**Impact Assessment**

- Network bandwidth saturation observed
- Web server response time increased significantly
- Service downtime recorded for approximately 15 minutes

**Indicators of Compromise (IOCs)**

- Multiple connection attempts from unknown external IPs
- Abnormal spike in TCP SYN requests
- Repeated HTTP GET requests from distributed sources

**Response Actions Taken**

- Suspicious IP addresses were blocked using firewall rules
- Rate limiting was enabled on network gateway
- Traffic filtering policies were implemented

**Remediation Measures**

- Network traffic monitoring enhanced
- Intrusion detection alerts configured
- System logs reviewed for further anomalies

**Lessons Learned**

Implementation of proactive monitoring tools and incident response procedures helped minimize service disruption and ensured timely mitigation of the attack.

**Prepared By: Pal Piyushkumar Patel**

**Date: 27-02-2026**

*Image Source: Self-generated Lab Work*

# Conclusion

The SOC concepts explored in this report were implemented within a virtual lab environment to gain structured understanding of monitoring techniques, log analysis, incident response workflows, and security tool utilization. The included screenshots serve as proof of task completion and demonstrate the successful implementation of SOC monitoring practices.

## References


- NIST Cybersecurity Framework
- MITRE ATT&CK Framework
- Elastic SIEM Documentation
- Splunk Security Monitoring Guide